BRAC UNIVERSITY

Inspiring Excellence

CSE461

Section : 8

Group : 5

Semester : Summer_2026

Robot Project Proposal

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Submission Date:11-7-2026

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Project Idea 1

1. Project Title: Hospital Medicine Delivery Robot

2. Purpose

Objective: To design an autonomous robot capable of delivering medicines to patient rooms inside a hospital without direct human contact.

Scope: Intended for use in hospitals, clinics, and nursing homes to transport prescribed medicines from the pharmacy/nurse station to designated patient rooms along a fixed route of checkpoints.

Significance: Reduces the physical workload on nursing staff and minimizes person-to-person contact, which is especially valuable during infectious disease outbreaks, lowering the risk of cross-infection between staff and patients.

3. Components

- **Microcontroller:** Arduino uno
- **Sensors:** Ultrasonic Sensor (HC-SR04) x2, IR Sensor x2, RFID Reader (RC522) with tags
- **Actuators:** DC Motors x4 (with wheels), Servo Motor (SG90) x1, Buzzer x1
- **Body/Chassis:** Custom robot chassis with a lockable medicine compartment
- **Additional Components:** 16x2 I2C LCD Display, L298N Motor Driver, 12V Li-ion Battery Pack, Jumper wires & breadboard

4. Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino uno	1200	1	1200
Ultrasonic Sensor (HC-SR04)	120	2	240
IR Sensor	100	2	200
RFID Reader (RC522) + Tags	350	1	350
DC Motor with Wheel	250	4	1000
Servo Motor (SG90)	250	1	250
Buzzer	30	1	30
L298N Motor Driver	180	1	180
Robot Chassis (with compartment)	900	1	900
16x2 I2C LCD Display	350	1	350
12V Li-ion Battery Pack	700	1	700
Jumper Wires & Breadboard	200	1	200
		Total	**5600 BDT**

5. Functionality Breakdown

Functionality 1: Checkpoint-Based Navigation using RFID

Overview: The robot follows a predefined hospital route and identifies each ward/room checkpoint using RFID tags placed along the corridor.

Working Procedure: RFID tags are mounted at fixed points (e.g., nurse station, ward entrances). As the robot moves along the route, the RC522 reader scans nearby tags; the Arduino uno matches the scanned tag ID against the delivery destination stored for the current task and decides whether to continue, turn, or stop at that checkpoint. The LCD displays the current status (e.g., 'En route to Ward 3').

Functionality 2: Obstacle Avoidance using Ultrasonic Sensor

Overview: The robot detects and avoids obstacles such as people, carts, or equipment while moving through hospital corridors.

Working Procedure: The front-facing ultrasonic sensor continuously measures the distance to obstacles ahead. If an object is detected within a threshold distance, the Arduino halts the DC motors, sounds the buzzer as a warning, and re-routes (e.g., stops and waits, or steers around the obstacle using the IR sensors for side clearance) before resuming the path to the checkpoint.

Functionality 3: Automated Medicine Compartment Unlocking

Overview: Once the robot reaches its destination, it automatically unlocks the medicine compartment for the staff or patient to retrieve the delivery.

Working Procedure: Upon confirming arrival at the correct RFID checkpoint, the Arduino triggers the servo motor to rotate and release the compartment latch. The buzzer sounds to alert staff, and the LCD shows a 'Delivery Ready' message. After a short timeout or a manual confirmation button press, the servo re-locks the compartment and the robot returns to the pharmacy station.

Project Idea 2

1. Project Title: Autonomous Warehouse Inventory Robot

2. Purpose

Objective: To build an autonomous robot that navigates warehouse aisles and inspects shelves to track and verify inventory.

Scope: Intended for use in warehouses and storage facilities to reduce the time and manual labor required for stock inspection and inventory verification.

Significance: Automating shelf inspection reduces the time needed for manual inventory checks, minimizes human counting errors, and allows staff to be reallocated to higher-value tasks.

3. Components

- **Microcontroller:** Raspberry Pi 4
- **Sensors:** RFID Reader (RC522), Ultrasonic Sensor (HC-SR04) x2, Camera Module
- **Actuators:** DC Motors x4 (with wheels), Servo Motor x1
- **Body/Chassis:** Wheeled robot chassis with elevated camera mount
- **Additional Components:** L298N Motor Driver, 12V Battery Pack, Jumper wires & breadboard

4. Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
Raspberry Pi 4 (4GB)	6500	1	6500
Camera Module	1200	1	1200
RFID Reader (RC522) + Tags	350	1	350
Ultrasonic Sensor (HC-SR04)	120	2	240
DC Motor with Wheel	250	4	1000
Servo Motor	250	1	250
L298N Motor Driver	180	1	180
Robot Chassis	900	1	900
12V Battery Pack	700	1	700
Jumper Wires & Breadboard	200	1	200
		Total	**11520 BDT**

5. Functionality Breakdown

Functionality 1: Autonomous Aisle Navigation

Overview: The robot moves through warehouse aisles on its own, following a mapped path between shelving units.

Working Procedure: The Raspberry Pi runs a navigation routine that drives the DC motors along a predefined aisle sequence, using the ultrasonic sensors to keep a safe distance from shelves on either side. When an aisle end or turning point is detected, the robot pivots and proceeds to the next aisle.

Functionality 2: RFID-Based Shelf Scanning

Overview: As it passes each shelf, the robot scans RFID tags attached to stored items to verify their presence and location.

Working Procedure: The RFID reader continuously scans for tags within range as the robot moves. Detected tag IDs are timestamped and logged by the Raspberry Pi along with the current aisle/shelf position, building a live inventory record that can be compared against the expected stock list.

Functionality 3: Visual Inventory Capture

Overview: The robot captures images of shelves to provide a visual record for stock verification and to catch items without functioning RFID tags.

Working Procedure: At each shelf position, the camera module captures an image, which is processed on the Raspberry Pi (or uploaded to a server) for basic image analysis such as detecting empty slots or misplaced items. Captured images are stored with metadata (aisle, shelf, timestamp) for later review by warehouse staff.

Project Idea 3

1. Project Title: Autonomous Smart Library Book Assistant Robot

2. Purpose

Objective: To design an autonomous robot that transports books between library shelves and the circulation desk or reading area.

Scope: Intended for use in libraries to assist with retrieving and delivering books, navigating between shelf rows and reading/service areas.

Significance: Reduces librarian workload for book retrieval and delivery, speeds up service for patrons, and can assist visitors with mobility difficulties.

3. Components

- **Microcontroller:** Arduino uno
- **Sensors:** RFID Reader (RC522), Ultrasonic Sensor (HC-SR04) x2, IR Sensor x2
- **Actuators:** DC Motors x4 (with wheels), Servo Motor x1
- **Body/Chassis:** Robot chassis with an open book-carrier tray/shelf attachment
- **Additional Components:** L298N Motor Driver, 12V Battery Pack, Jumper wires & breadboard

4. Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino uno	1200	1	1200
RFID Reader (RC522) + Tags	350	1	350
Ultrasonic Sensor (HC-SR04)	120	2	240
IR Sensor	100	2	200
DC Motor with Wheel	250	4	1000
Servo Motor	250	1	250
L298N Motor Driver	180	1	180
Robot Chassis	900	1	900
Book Carrier Tray Attachment	400	1	400
12V Battery Pack	700	1	700
Jumper Wires & Breadboard	200	1	200
		Total	**5620 BDT**

5. Functionality Breakdown

Functionality 1: Shelf Row Navigation

Overview: The robot navigates between library shelf rows to reach the location of a requested book.

Working Procedure: Each shelf row is tagged with an RFID marker. The Arduino drives the robot along the main aisle using ultrasonic distance readings to stay centered, and the RFID reader confirms which row the robot is passing so it can stop precisely at the row containing the requested book.

Functionality 2: Book Delivery to Desk/Reading Area

Overview: After a book is placed on the robot's carrier tray, the robot delivers it to the circulation desk or the requesting patron's reading area.

Working Procedure: Once loading is confirmed (via a button press or weight/IR check on the tray), the Arduino computes the return path to the destination checkpoint and drives the DC motors accordingly, using the same RFID checkpoints to confirm arrival before signalling completion (e.g., via buzzer/LED).

Functionality 3: Obstacle Avoidance

Overview: The robot avoids patrons, chairs, and other obstacles while moving through the library floor.

Working Procedure: The IR sensors and ultrasonic sensor continuously monitor the area immediately ahead and to the sides of the robot. When an obstruction is detected, the Arduino stops forward motion, briefly reverses or steers around the obstacle if there is clearance, and resumes its route once the path is clear.